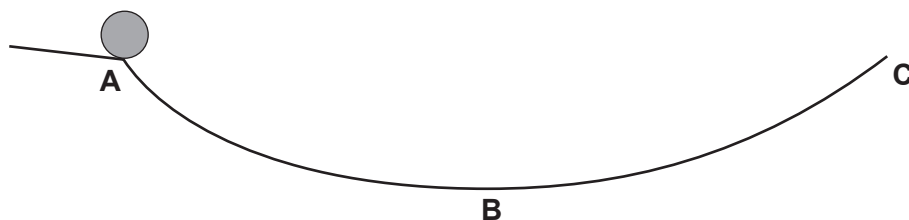


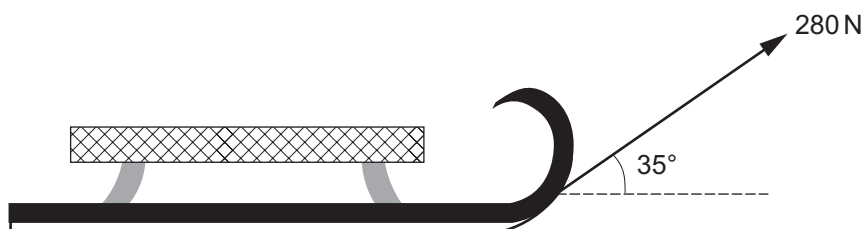
5. (a) The diagram shows a marble about to be released at **A** on a curved ramp.



Explain the energy transfers that take place from the moment the marble is released to the moment it finally comes to rest. [6 QER]



- (b) A heavy sled moves with **constant velocity** when it is pulled by a force of 280 N acting as shown.



- (i) State why the work done in pulling the sled **cannot** be calculated by simply multiplying 280 N by the distance the sled is pulled. [1]

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- (ii) It takes 20 minutes to pull the sled a distance of 3.0 km over level ground. Calculate the mean power needed. [4]

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